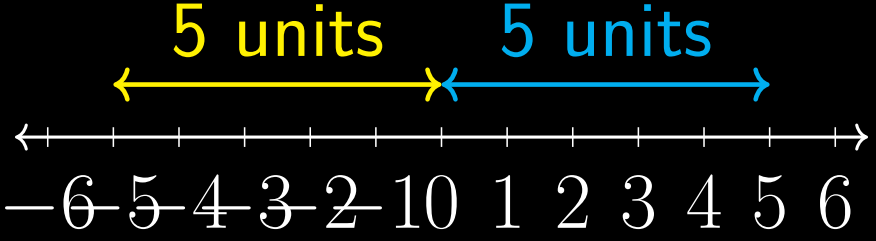
Absolute Value of Integers

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Absolute Value is the distance of an integer from zero on the number line, regardless of direction.

Absolute Value on the Number Line



Both -5 and 5 are 5 units from zero.

Absolute Value Notation

The symbol for absolute value is two vertical bars placed around an integer.

$$|-5|$$

This is read as "the absolute value of -5."

Distance from Zero

Distance from zero is always non-negative.

Therefore, the absolute value of an integer is always zero or positive.

$$|-5| = 5 \qquad |5| = 5$$

Absolute Value of Zero

The **absolute value of zero** is zero because zero is zero units away from itself.

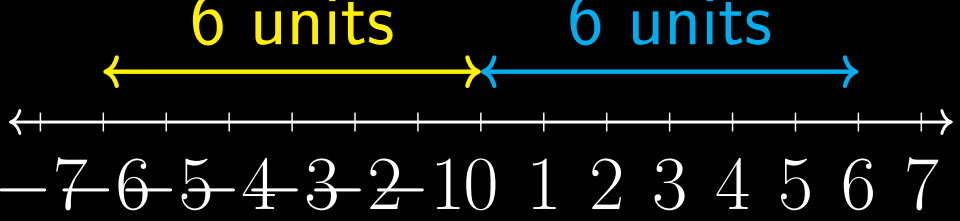
$$|0| = 0$$

Opposite Integers

Opposite integers have the same absolute value because they are equally distant from zero but lie on opposite sides of the number line.

$$|6| = |-6| = 6$$

Opposites on the Number Line



Same distance from zero, so the same absolute value.

Never Negative

The absolute value of an integer is never negative.

$$|-8| = 8, \text{ not } -8$$

Memory Tip

Absolute value = **A**way from zero

It tells us **how far**, not **which way**.

Summary Table

Integer	Absolute Value
-9	9
-3	3
0	0
3	3
9	9

Absolute Value Is Distance

- **Positive:** The absolute value is the integer itself.
- **Zero:** The absolute value is zero.
- **Negative:** The absolute value is its opposite (positive).

$$|4| = 4 \qquad |0| = 0 \qquad |-4| = 4$$

Examples

Example 1

Find $|-9|$.

$$|-9| = 9$$

because -9 is 9 units from zero.

Example 2

Find $|12|$.

$$|12| = 12$$

because 12 is 12 units from zero.

Example 3

Find $|0|$.

$$|0| = 0$$

because 0 is 0 units from itself.

Example 4

A diver is 4 meters below sea level.

$$\text{Position} = -4$$

$$|-4| = 4$$

The diver is 4 meters away from sea level.

Practice Exercises

Directions

Write the absolute value of each integer or respond to the question about its position or distance on the number line.

Practice 1

Find the absolute value of -7. Write your answer using absolute value notation.

Practice 2

Find the absolute value of 9. Write your answer using absolute value notation.

Practice 3

What is the absolute value of 0?

Practice 4

Which integer has an absolute value of 6: -6 or 6? Explain briefly.

Practice 5

On a number line, how many units away from zero is the integer -8?

Practice 6

On a number line, how many units away from zero is the integer 5?

Practice 7

Complete the statement: $|-12| =$ _____.

Practice 8

Complete the statement: $|3| =$ _____.

Practice 9

Two points on a number line represent -10 and 10. What do you notice about their absolute values?

Practice 10

A submarine is 15 meters below sea level. Represent its position using an integer and state the absolute value of that integer. What does the absolute value mean in this situation?

Activities

Directions

Write the absolute value of each integer or respond to the question about its position or distance on the number line.

Activity 1

Find the absolute value of -11. Write your answer using absolute value notation.

Activity 2

Find the absolute value of 13. Write your answer using absolute value notation.

Activity 3

What is the absolute value of 0?

Activity 4

Which integer has an absolute value of 9: -9 or 9? Explain briefly.

Activity 5

On a number line, how many units away from zero is the integer -14?

Activity 6

On a number line, how many units away from zero is the integer 7?

Activity 7

Complete the statement: $|-16| =$ _____.

Activity 8

Complete the statement: $|5| =$ _____.

Activity 9

Two points on a number line represent -12 and 12. What do you notice about their absolute values?

Activity 10

A diver is 18 meters below sea level. Represent the diver's position using an integer and state the absolute value of that integer. What does the absolute value mean in this situation?

Thank you!